

CSC1005/CSC1022 CA2 Assignment

ER Modelling

Goal: You have been given a company/organisation and you need to design an Entity Relationship Model for them.

This should be a 2 page report, the first of which should contain the ER Diagram in detail and the next page should contain all the explanations/answers of the tasks below.

Any resources or external references, such as websites or publications, utilised during the design process must be appropriately cited.

Task 1:

Design an ER diagram for the organisation/company which has been assigned to you. You should have at least 5 entities.

Consider between 3 and 5 simple attributes for each entity.

2 of the entities should have complex attributes and 1 entity should have a multi-valued attribute.

1 of the entities should have a derived attribute.

Mention in detail why each attribute is selected as its respective type.

Task 2:

Nominate one of the 5 entities as a weak entity.

Explain:

Why it is a weak entity and which entity is the owner

What type of participation constraint is used

What is used as the partial key for this entity

Task 3:

Include one example of each of the following relationships: one-to-one, one-to-many, and many-to-many. Add any other remaining relationships you want.

Explain why you gave each relationship its cardinality ratio.

Did you have any other participation constraints?

Task 4:

Convert the ER diagram to a Relational Model using the ER Mapping steps.
